Semester	B.E. Semester VII – Computer Engineering
Subject	Scalable ML & BDA Lab
Subject In-charge	Prof. Shweta Jaiswar

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<table>
<tr><td>Experiment Number</td><td colspan="2">04</td></tr>
<tr><td>Experiment Title</td><td colspan="2">Dimensionality Reduction using Principal Component Analysis-PCA</td></tr>
<tr><td>Resources / Apparatus Required</td><td>Hardware:</td><td>Software:</td></tr>
<tr><td>Description</td><td colspan="2">The objective of this experiment is to apply Principal Component Analysis (PCA) for dimensionality reduction using PySpark MLlib. The COVID-19 tweet dataset was obtained using KaggleHub and converted from a Pandas DataFrame into a PySpark DataFrame. Since PCA requires numerical input, numerical features such as tweet length, word count, hashtag count, mention count, URL count, exclamation count, question count, and uppercase character count were extracted from the tweet text.

The extracted features were combined into feature vectors using VectorAssembler and standardized using StandardScaler. PCA was then applied to reduce the original 8-dimensional feature space to 2 principal components. The reduced feature space was visualized using a 2D scatter plot, and the original and reduced feature spaces were compared to understand the effect of dimensionality reduction.</td></tr>
<tr><td>Observation</td><td colspan="2">The dataset contained 41,157 records and 6 original columns. Eight numerical features were derived from the OriginalTweet column and used for PCA. After feature vectorization, the features were standardized using StandardScaler so that features with different scales could be compared fairly.
PCA reduced the feature space from 8 dimensions to 2 dimensions. The first principal component explained 26.59% of the total variance, while the second principal component explained 19.12%. Together, the two components preserved 45.71% of the total variance.
The PCA scatter plot provided a two-dimensional representation of the dataset, making it easier to visualize the distribution of the observations.</td></tr>
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Compared with the original 8-dimensional feature space, the PCA representation is considerably simpler, although some information is lost during the reduction.

Program

PCA_Exp_SML

```
PySpark DataFrame:
+--------+----------+---------+----------+--------------------------------------------------------------------+-------------
|UserName|ScreenName| Location|   TweetAt|                                                       OriginalTweet|        Senti
+--------+----------+---------+----------+--------------------------------------------------------------------+-------------
|    3799|     48751|   London|16-03-2020|@MeNyrbie @Phil_Gahan @Chrisitv https://t.co/iFz9FAn2Pa a...|          Neu
|    3800|     48752|       UK|16-03-2020|advice Talk to your neighbours family to exchange phone n...|         Posi
|    3801|     48753|Vagabonds|16-03-2020|Coronavirus Australia: Woolworths to give elderly, disabl...|         Posi
|    3802|     48754|      NaN|16-03-2020|My food stock is not the only one which is empty...\r\r\n...|         Posi
|    3803|     48755|      NaN|16-03-2020|Me, ready to go at supermarket during the #COVID19 outbre...|Extremely Nega
+--------+----------+---------+----------+--------------------------------------------------------------------+-------------
only showing top 5 rows

Numerical Features:
+------------+----------+-------------+-------------+---------+-----------------+--------------+---------------+
|tweet_length|word_count|hashtag_count|mention_count|url_count|exclamation_count|question_count|uppercase_count|
+------------+----------+-------------+-------------+---------+-----------------+--------------+---------------+
|         111|         8|            0|            3|       24|                0|             0|             17|
|         237|        38|            0|            0|        0|                0|             0|              3|
|         131|        14|            0|            0|        8|                0|             0|             13|
|         306|        42|            7|            0|        8|                0|             0|             60|
|         310|        40|            6|            0|        8|                0|             0|             16|
+------------+----------+-------------+-------------+---------+-----------------+--------------+---------------+
only showing top 5 rows
```

PySpark DF and Numerical Features

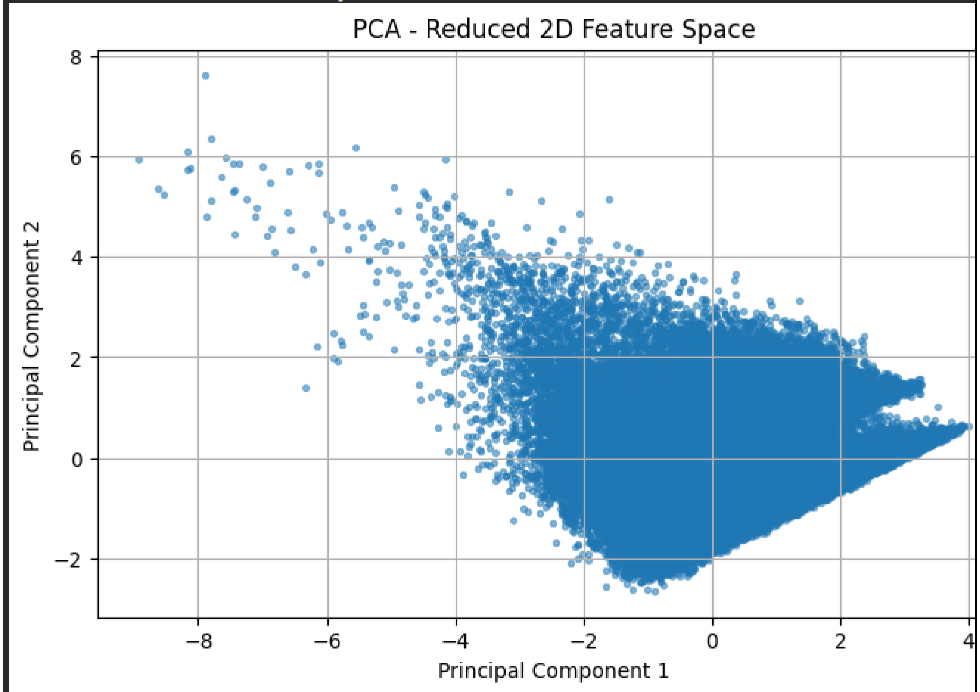
```
Feature Vectors:
+-------------------------------------+
|raw_features                         |
+-------------------------------------+
|[111.0,8.0,0.0,3.0,24.0,0.0,0.0,17.0]|
|(8,[0,1,7],[237.0,38.0,3.0])         |
|(8,[0,1,4,7],[131.0,14.0,8.0,13.0])  |
|[306.0,42.0,7.0,0.0,8.0,0.0,0.0,60.0]|
|[310.0,40.0,6.0,0.0,8.0,0.0,0.0,16.0]|
+-------------------------------------+
only showing top 5 rows
```

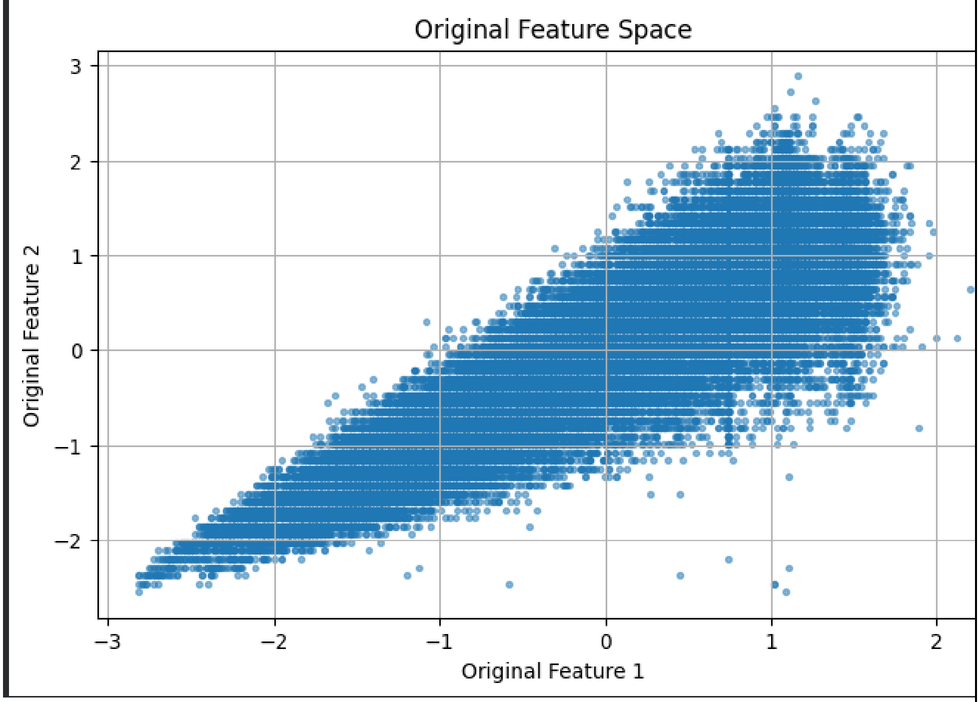
Feature Vector

```
Standardized Features:
+------------------------------------------------------------
|scaled_features
+------------------------------------------------------------
|[-1.3575119814502183,-1.9405807454341424,-0.69416045829658
|[0.477747839983598,0.6479825721243515,-0.6941604582965875,
|[-1.0662008986829459,-1.4228680819224437,-0.69416045829658
|[1.482771075530688,0.9931243477988173,1.7593659050304715,-
|[1.5410332920841425,0.8205534599615845,1.4088621388408917,
+------------------------------------------------------------
only showing top 5 rows

PCA Reduced Features:
+-----------------------------------------+
|pca_features                             |
+-----------------------------------------+
|[0.8954862679603238,3.129852865856212]   |
|[0.16479464325847526,-1.593293709981892]|
|[1.5922621191229103,1.0016088781423893] |
|[-3.6112131761948785,1.9857580230223035]|
|[-1.9215344315662448,0.3340637575144435]|
+-----------------------------------------+
only showing top 5 rows

Explained Variance:
PC1: 26.59%
PC2: 19.12%
```

Standard scaling those vectors and then finally using PCA reduction on it

```
Total Information Preserved by 2 PCs: 45.71%
```





Data Visualization between original and PCA

Original vs Reduced Feature Space:

	Feature Space	Number of Dimensions	Information Preserved
0	Original	8	100%
1	PCA Reduced	2	45.71%

Information Preservation

Conclusion:	The experiment successfully demonstrated **dimensionality reduction using PCA with PySpark MLlib**. The original 8 numerical features were transformed into 2 principal components, providing a simpler representation of the dataset. The first two components retained **45.71% of the total variance**, showing that they captured a significant portion of the variation present in the original features. Although reducing the data to two dimensions resulted in some information loss, it made the high-dimensional dataset easier to visualize and analyze. Therefore, PCA can be useful for **simplifying feature spaces, reducing computational complexity, and visualizing high-dimensional data** in scalable machine-learning applications.